

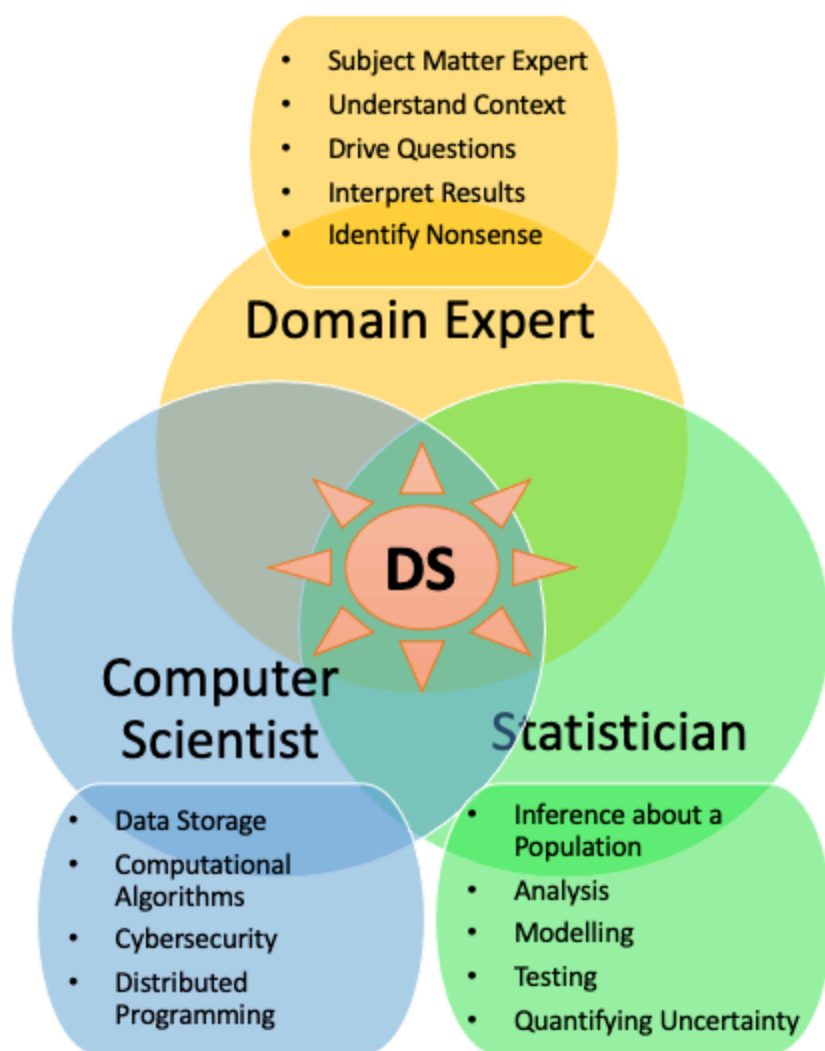
1.0 Data Science and R

Data science blends statistics, computing, and a domain question. This course uses R for that cycle.

1. What this course is for

DATA 612 is statistical programming in R, inside a data-science workflow. You will import data, tidy it, plot it, summarize it, and write the work so someone else (or you, next month) can rerun it.

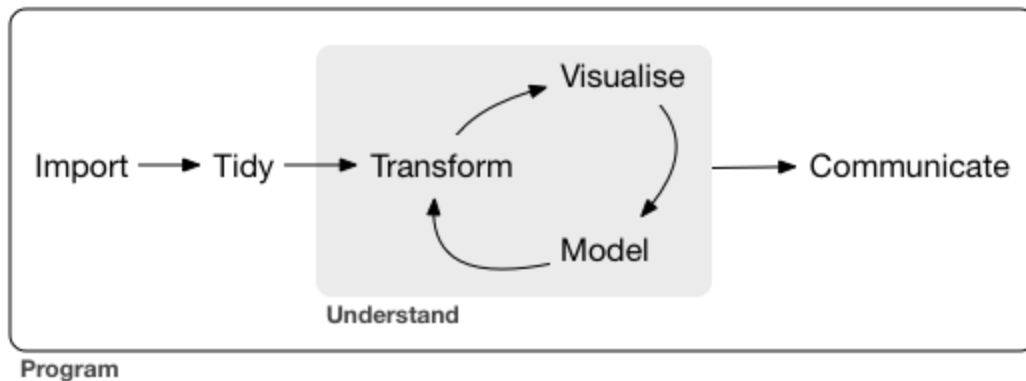
Data science is a blend of statistics, computing, and a question from a domain. No one person owns every piece. A project is usually a team.



R is the language for this course because it was built for statistics, it is free, and the tidyverse is a consistent grammar for tables. Python is also widely used. We stay with R.

2. A working cycle

Wickham and Grolemund's picture in *R for Data Science* is the one we will keep coming back to: import, tidy, transform, visualize, model, communicate, then go around again.



The first pass is almost never the last. A plot changes the question. A model sends you back to the table.

3. Tools for the cycle

Job	What we use
Language	R
Editor	RStudio / Posit Desktop
Tables	tidyverse: <code>dplyr</code> , <code>tidyr</code> , <code>readr</code>
Plots	<code>ggplot2</code>
Documents	Quarto (or R Markdown)

The syllabus on the course page is the contract for due dates and policies. Chapter 1 of *R for Data Science* is the reading that matches this note.

4. Practice

1. In one sentence, what mix of skills does a data-science project need?
2. Name the six verbs on the R for Data Science cycle.
3. Why does this course use R rather than a spreadsheet for the same tables?